



Smart Buildings

Energy Efficiency &
Occupant Experience

CLIMORA

Smarter Cooling from Condensate

Climate-aware **L**oad **I**ntelligence & **M**oisture **O**ptimization
for Resource-efficient Air-conditioning



Team

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FULL REPORT & CODE

github.com/adityakalagatoori/YUVA-CLIMORA



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THE PROBLEM

Cooling Energy Is Large.
Visibility Is Low.

India's commercial buildings consume a large share of electricity for cooling, yet most buildings lack real-time visibility and discard AC condensate.

01

A major energy load



33%

of India's electricity is consumed by buildings



~40%

of commercial-building energy use is HVAC / cooling

Sources: Yuva Yodha Tech Hackathon 2026;
BEE / ECBC Users' Manual

02

Limited visibility

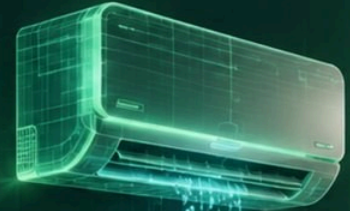


Real-time data often unavailable.

Many existing buildings are not equipped with detailed building-management systems, leaving owners and facility managers with limited equipment-level visibility.

03

A continuously discarded resource



Condensate water is produced by every operating AC.



Today:
Drained away.



High cooling demand + low visibility + discarded condensate =
an overlooked efficiency opportunity.



THE OPPORTUNITY

An AC Already Produces
a Usable Resource.

Warm
humid
air



Indoor Unit



Condensate
water



Outdoor Unit (Condenser)

Heat
rejected
to outside

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01

AC removes moisture
from indoor air

Every AC removes moisture from indoor air to cool it. This moisture condenses into liquid water at the indoor coil.



Near-distilled water
(very low mineral content)

02

Today, it is discarded

The condensate is routed straight to a drain pipe and discarded, in every AC, in every building, continuously — with zero reuse mechanism in current practice.



Drain

03

It can help cool the condenser

Spraying this water as a fine mist across the outdoor condenser (adiabatic pre-cooling) lowers the effective inlet air temperature, reducing the heat-rejection burden on the compressor — a well-established HVAC technique, demonstrated using AC condensate in a 2025 study.



The water an AC throws away could help **cool**
the **very machine that produced it.**



CLIMORA

From Condensate to Controlled Pre-Cooling

A simple retrofit that captures AC condensate and uses it to cool the outdoor condenser through a fine water mist.



Indoor Unit
Produces condensate while cooling

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5

Outdoor Condenser

Mist cools the inlet air before it reaches the coil

2

T-Junction

Diverts a portion of condensate

3

Collection Tank (on load cell)

Stores captured condensate

4

Ultrasonic Atomizer

Converts water into fine mist

01

AC produces condensate



Moisture from indoor air condenses into water.

02

T-junction diverts a portion



A T-junction diverts some water while the original drain remains **fully open**.

03

Collection tank (on load cell)



Captured condensate is stored in a covered tank placed on a load cell.

04

Ultrasonic atomizer



Turns the stored water into a fine mist.

05

Mist cools condenser



Mist is sprayed at the outdoor condenser inlet, lowering the air temperature and reducing the AC's electrical work.



CLIMORA adds one new path alongside the AC's existing drain – it **never replaces it**.



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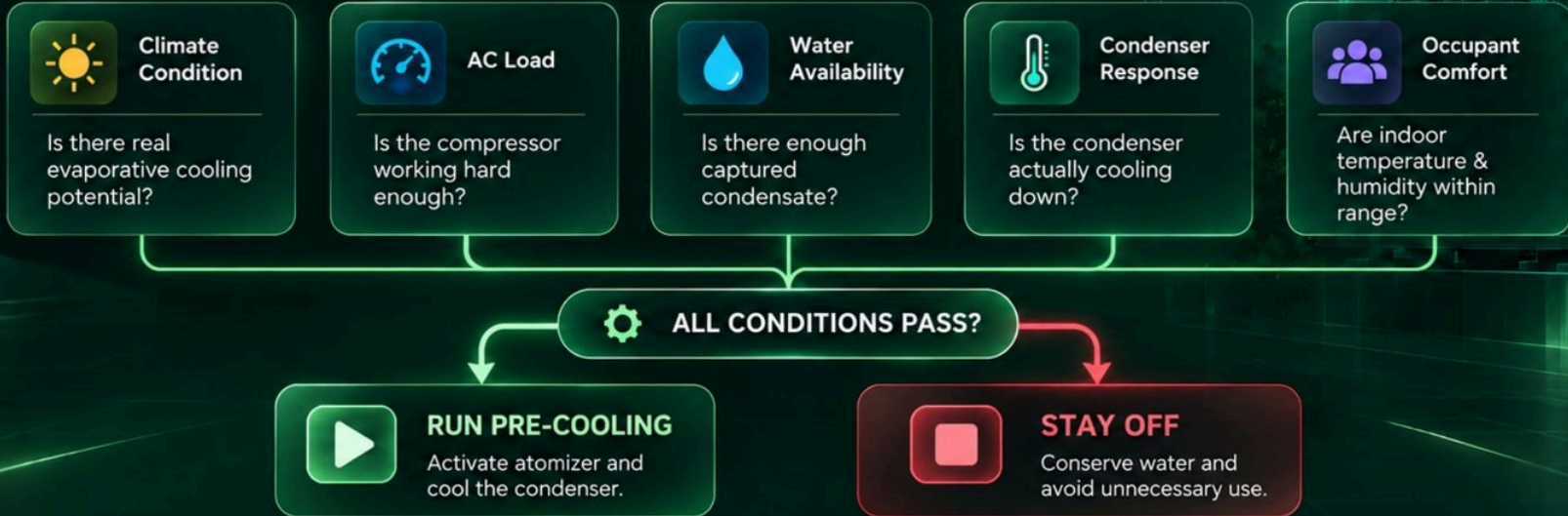
CLICK HERE



THE DIFFERENCE

Decides When **NOT** to Run.

The mechanism is known. **We add the decision layer.**





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SYSTEM ARCHITECTURE

From AC Sensor to Building Dashboard



One AC node today.

The same architecture scales to a building gateway tomorrow — nothing about the sensing or control layer has to change.

Technologies:



ESP32



MQTT (Mosquitto)

PostgreSQL / TimescaleDB
(Local)

React (Dashboard)



Mock API (BMS-ready)



RETROFIT DESIGN

Low-Cost, Fail-Safe and AC-Agnostic

1 Low-Cost



- Estimated prototype cost: ₹7,000 – ₹8,500
- Built from off-the-shelf components (ESP32, energy meter, sensors, atomizer, etc.)
- No custom-fabricated parts required.

2 Fail-Safe by Design



- Original AC drain is never blocked (we branch from it).
- If tank is full → overflows to original drain (gravity-fed).
- If controller/sensors/network fail → pre-cooling stays OFF, drainage unaffected.
- Pre-cooling is suspended if indoor conditions go out of range (comfort first).

3 Easy to Install



- Only touches the drain line and outdoor unit surroundings.
- Refrigeration circuit is never opened or modified.
- Install sequence:



Tap
drain line



Mount tank
+ sensors



Position
atomizer



Calibrate
& run

4 Starts Narrow, Scales Deliberately



- Initial target: split and cassette air-cooled AC units (most common in Indian commercial buildings).
- Designed to extend to packaged/ducted systems.
- Future roadmap: larger centralized/VRF systems via building-gateway architecture.



Touches a drain line, not a refrigeration circuit — the lowest-risk way to retrofit an existing building.

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Atomizer
(mist at inlet)

T-junction
(from drain line)

Collection Tank
(on load cell)

ESP32
(controller)

Target AC Types (Phase 1)



Split AC



Cassette AC



Package/Ducted
(Future)



Centralized / VRF
(Future)



HOW WE PROVE THE ENERGY SAVING

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1

Net Energy Saving (What We Report)

We report net energy saving — the AC's saving minus the electricity γ used by the atomizer and control electronics themselves.



$$\text{Net Energy Saving (percentage)} = \frac{E_{ac, \text{baseline}} - (E_{ac, \text{intervention}} + E_{aux})}{E_{ac, \text{baseline}}} \times 100$$

 $E_{ac, \text{baseline}}$ AC energy without
pre-cooling (kWh) $E_{ac, \text{intervention}}$ AC energy with
pre-cooling (kWh) E_{aux} Atomizer +
controller energy (kWh)

2

Controlled A/B/A/B Testing (How We Measure)

Baseline
(45–60 min)Intervention
(45–60 min)Baseline
(45–60 min)Intervention
(45–60 min)Same AC, same room,
same setpoint

Same fan setting

Doors/windows
closedRepeat on at least two γ different days/times, γ γ .If indoor temperature/
humidity drifts outside the
allowed range, the trial is discarded.

3

Where the Numbers Stand (Kept Separate)

Tier	Value	Status
 Literature result (published study)	~39%	Someone else's result, cited as context only.
 Our target (CLIMORA)	≥5%	Our conservative goal.



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IMPACT: ENERGY, WATER, CARBON AND COMFORT

Four measurable outcomes, one framework per AC node.

1



ENERGY

Net electricity saved

$$\text{Net Energy Saving} = E_{ac, \text{baseline}} - (E_{ac, \text{intervention}} + E_{aux})$$

(kWh or %)

2



WATER

Condensate captured & reused

Measured via load cell
Water reused = m_{captured}
(grams or litres)

3



CARBON

CO₂e avoided

CO₂e avoided =
(Net kWh saved) × 0.705
0.705 kgCO₂/kWh
(CEA grid emission factor, FY2025-26)

4



COMFORT

% time within comfort boundary

Indoor temperature & humidity monitored continuously
Pre-cooling suspends automatically on violation (Slide 5).



Illustrative Scaling & Impact (Per AC Unit)



1.5 – 2 ton
commercial AC
(typical commercial use)



~3,150 kWh/year
baseline energy
(at typical operating hours)



~₹1,400 – 2,800/year
saved per AC unit
(at 5–10% net saving target)



~1.1 – 2.2 tCO₂/year
avoided per AC unit
(at 5–10% net saving target)



~100 – 300 litres/year
reused per AC unit
(varies by climate & usage)



Scales linearly with AC count
Same per-unit model for 10 units or 100 units.



Every AC node delivers measurable value:

energy saved, water reused, CO₂ avoided, and comfort preserved — and the impact grows linearly with scale.

FROM ONE AC TO A SMART BUILDING

Scaling, Climate Fit & Economics

1

SCALING ARCHITECTURE

Same node, from 1 AC to 100+ units.



Each AC gets one CLIMORA node (sensors + ESP32).
Only the aggregation layer grows.

2

CLIMATE FIT IN INDIA

Where it works best and why.



Hot-Dry

High evaporative potential, limited condensate.



Warm-Humid

High condensate, limited evaporative potential.



Composite (Delhi-like)

Best fit — seasonal balance of both.



Temperate / Cold

Mostly HOLD.



Activates only when water + cooling conditions + AC load justify pre-cooling —not a one-size-fits-all approach.

3

ECONOMICS (ILLUSTRATIVE, SCALES BY AC COUNT)

Same per-unit model, larger buildings scale linearly.



Small Office
(10 AC units)



₹40,000 – 80,000
Investment (illustrative)



~₹28,000 / year
Savings @ 10% target



~1.4 – 2.8 years
Payback (illustrative)



Mid-size Hotel
(50 AC units)



₹2 – 4 lakh
Investment (illustrative)



~₹1.4 lakh / year
Savings @ 10% target



~1.4 – 2.8 years
Payback (illustrative)



Large Commercial
(100 AC units)



₹4 – 8 lakh
Investment (illustrative)



~₹2.8 lakh / year
Savings @ 10% target



~1.4 – 2.8 years
Payback (illustrative)

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THE SAME NODE, THE SAME LOGIC — FROM A SINGLE AC TO A FULL BUILDING.

CLIMORA scales through aggregation, not complexity.



YUVA YODHA TECH HACKATHON 2026

IMPLEMENTATION ROADMAP & TEAM

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PROJECT ROADMAP

This Hackathon (Phase 1 – 4)

Future (Phase 5 – 6)

1



**Bench
Validation**

Sensor/actuator
checks, decision
logic (no live AC).



Build-ready

2



**Single AC
Prototype**

Full physical
integration
on one unit.



Build-ready

3



**Controlled
Testing**

A/B/A/B
energy-saving
measurement



Build-ready

4



**Multi-condition
Testing**

Different times
and weather
conditions.



Build-ready

5



**Building
Pilot**

Multiple AC nodes
+ gateway
aggregation



Future

6



**BMS / Grid
Integration**

Real API integration
and demand-response
(via DISCOM programmes)



Future



OUR APPROACH



1. BUILD FIRST

Physical retrofit + sensing +
decision engine on a single AC.



2. VALIDATE NEXT

Real, measured net energy savings
through controlled testing.



3. SCALE LATER

Multi-AC gateway, building dashboard,
and real grid-signal integration.



“We know exactly what’s built, what’s being tested, and what’s planned — and we’re not blurring the three.”

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THANK YOU

SMART COOLING FOR A GREENER TOMORROW



Vaishnav Aditya K

SYSTEMS LEAD

Architecture • Hardware Integration
Control Logic • Deployment Planning



Balachandar A

SOFTWARE LEAD

Firmware & IoT Software
Gateway • Dashboard • BMS Interface



Shyaam Ganesh

DATA & ANALYTICS LEAD

Data Collection • A/B Testing
Energy Analysis • Impact Modelling



Srushti D

PRODUCT & DEPLOYMENT LEAD

Mechanical Design • Enclosure
UI/UX • Installation • Field Testing



FULL REPORT & CODE

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